

Leo Zhang

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PROFESSIONAL SUMMARY

Senior **Android** Architect with **10+ years** building large-scale **Android** platforms, real-time messaging systems and **mobile** infrastructure using modern **Android technologies**. Senior Software Developer - **Android** — Expert in Kotlin, Java, **NDK/JNI**, **Android** architecture, Flutter, cross-platform **mobile** development and web **technologies** including and TypeScript. Designed and delivered high-**quality Android** systems supporting 5,000+ DAU and 500K+ daily messages, including messaging platforms, **mobile** SDKs, hotfix frameworks and performance engineering tooling with ongoing enhancements and extension. Strong background in WebSocket, TCP/UDP, cold start optimization, ANR/OOM resolution, CI/CD automation and AI-powered applications published at IVCNZ 2025. Master's in Computer and Information Sciences from AUT, **First Class Honours**.

CORE COMPETENCIES

Android & Native: Kotlin, Java, **Android** SDK, Material Design

Real-time Messaging: WebSocket, TCP/UDP, IM Architecture, Offline Sync

Performance: Cold Start, Memory & OOM, ANR Resolution, APK Size

Mobile Infrastructure: Multi-Module Architecture, DevOps, SDK Development, CI/CD

Cross-Platform & Web: Flutter, TypeScript, JavaScript, Responsive UI/UX

Engineering Practices: Agile / Scrum, Mentoring, Code Review, Automated Testing

AI & Tools: PyTorch, Computer Vision, LLM, RAG

EXPERIENCE

Auckland University of Technology — *Auckland, New Zealand*

2024 – 2026

AI Research & Mobile Engineer

- Designed and delivered ChatClothes, a digital multimodal AI platform integrating computer vision, diffusion models, and LLM-powered workflows with RAG, vector search, and prompt grounding. Conducted extensive image dataset work, model training and testing across multiple CV projects. Provided technical leadership and owned the product end-to-end as cloud-native AI solutions, from system design and development through evaluation, deployment, and user validation.
- Deployed and optimized AI models on edge devices including Raspberry Pi and **Android** phones, running lightweight LLMs and vision models on resource-constrained ARM hardware with offline inference, image processing, and on-device prediction. Validated model outputs through ground truth testing and evaluation.

Tech: Python, PyTorch, Ollama, Docker, Linux, AWS, Azure, Git, Raspberry Pi, Android

Chunxiao Technology Co., Ltd. — *China*

2013 – 2024

Technology company (est. 2013, ~40 employees) delivering software, hardware, and system integration across streaming media, enterprise platforms, IoT, and industrial automation.

Senior Software Engineer | 2018 - 2024

- Smart Factory IoT Platform: architected manufacturing platform across 10+ factory sites with 6-person cross-functional team. Integrated RFID readers, barcode scanners, electronic scales (RS232/RS485 serial bridge to WebSocket), conveyors, and **Android** workshop terminals with Spring Cloud microservices. Event-driven architecture via ActiveMQ/Kafka, Docker deployment, 30%+ efficiency gain.

- IoT Device Management Platform: built **Android** device discovery/binding/remote control app and gateway firmware for smart switches and Zigbee/WiFi gateways. Protocol handling, MQTT cloud uplink, Spring Cloud backend API, and Vue.js management dashboard.
- Boobit Crypto Trading App: sole **Android** developer for **financial** cryptocurrency trading app. Real-time market data via WebSocket, order book rendering, candlestick charting, encrypted local wallet storage, MVVM with Clean Architecture, Retrofit/OkHttp, Room — delivering **banking**-grade reliability for sensitive **financial** transactions.
- Picture Book Locker: **Android** cabinet control system with electromagnetic locks, UV sterilization, sensors, face recognition + QR code authentication. Borrowing client app with <3s checkout time, 24/7 self-service.

Tech: Kotlin, Java, Android SDK, Spring Cloud, REST APIs, MySQL, Redis, Docker, CI/CD, WebSocket, NDK/JNI, MQTT, Zigbee, RFID

Senior Android Engineer | 2013 - 2018

- Enterprise Messaging Platform: architected **Android** client for 5,000+ DAU and 500K+ daily messages with sub-200ms latency. Built NDK/JNI TCP/UDP transport layer, WebSocket connection persistence, offline message sync, JPush backup push channel, and OEM ROM keep-alive strategies (Xiaomi/Huawei/OPPO). Migrated core IM from unstable in-house C++ to Easemob cloud, eliminating 90%+ defects. Built FastDFS distributed file storage (100GB+). Applied hotfix framework (Tinker, AndFix) for emergency patches and performance **quality** engineering across cold start, ANR, OOM, and APK size.
- Live Streaming Commerce System: full-stack **Android** + web **delivery** for live commerce platform. **Android** client in Java/Kotlin with RTMP/WebRTC/HLS streaming, multi-language architecture (C# + C++ media core + Python analytics + Lua config), 1000+ peak concurrency, 99.5% stream availability.

Tech: Kotlin, Java, Android SDK, NDK/JNI, C++, WebSocket, TCP/UDP, Gradle, CI/CD, Tinker, AndFix, JPush, RTMP, WebRTC, HLS, Ea

Additional Experience — *Auckland, New Zealand*

Customer Service & Operations (Part-time)

- Gilmours & Red Cross Shop: customer service, inventory management, and team collaboration

EDUCATION

Master of Computer and Information Sciences *Auckland University of Technology, NZ* 2024 – 2026
Research: Computer Vision, Diffusion Models, Large Language Models, Multimodal AI

Bachelor of Software Engineering *Hebei Univ. of Sci. & Tech., China* 2009 – 2013
Coursework: Data Structures, Algorithms, Java Programming, Database Systems, Software Engineering

LICENSES & CERTIFICATIONS

IoT Fundamentals — Xi'an Jiaotong University 2020
Science and Technology Achievement Award (Smart Manufacturing) — Hebei, China 2020
Software Design Engineer Certificate — China Qualification Authority 2019

INTERESTS

Hiking, self-hosted infrastructure, AI tools, and **mobile** technology trends

REFEREES

References available upon request.